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Challenges in FPGA Resource Allocation and Placement for Efficient EM Fault Attack Detection Strategies

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Abstract. The integration of FPGAs into security-critical systems entails a thorough evaluation of their vulnerability to fault injection attacks. ElectroMagnetic Fault Injection (EMFI) represents a particularly concerning threat vector due to its non-invasive characteristics, affordability, and capacity for precisely inducing transient faults. Our research examines the susceptibility to single EM pulsed attacks of Ring Oscillators (ROs), which can serve as a building block for both security primitives and EMFI sensors. By conducting different experiments on 28nm Xilinx FPGAs, we reveal that RO harmonic responses significantly vary based on the spatial configuration and routing constraints of the ring within the FPGA die. Additionally, we demonstrate the paramount importance of evaluating the EM pulse polarity and injection timing effects as they critically influence the fault distribution pattern while enabling a full control over the induced harmonic errors. These findings underscore the critical role of the FPGA resource utilization in security, providing valuable insights for the design of effective EMFI countermeasures and robust RO-based hardware implementations.

Keywords: Ring oscillators, Pulsed EM Fault Injection, FPGA, Place & Route, Harmonic locking.

1 Introduction

In today's digital landscape, ensuring the security of electronic systems is paramount, as they can be targeted by malicious entities seeking to compromise data confidentiality and system integrity. To this end, physical attacks and notably fault attacks have become a prevalent method for circumventing the security mechanisms of embedded devices, with a large arsenal of techniques at the attackers' disposal. These techniques range from disrupting clock signals to inducing sudden variations in temperature, supply voltage or substrate bias, along with injecting parasitic currents using powerful electromagnetic (EM) disturbances [1]. Other sophisticated techniques using laser or X-rays can be more efficient but very expensive and risky for the attackers as they can damage the target device. Among these methods, ElectroMagnetic Fault Injection (EMFI) has gained prominence due to the optimal balance between the attack effectiveness and its low cost and device preparation requirements. In a recent paper [2], an open-source design 'PicoEMP' was introduced whereby an EM pulsed fault injection

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can be mounted for less than 100$. On one hand, this makes EMFI attacks more accessible in both academia and industry, which will help improve our understanding of its fault mechanisms. On the other hand, this render the attack easier and more accessible for malicious entities as well, therefore extending the threat on several applications.

EMFI alters the behavior of a target device by inducing ground bounces or voltage drops as a result of the magnetic flux created by the EM probe [3]. The EM coupling can be performed either through harmonic or pulsed Fault Injection. In the context of our study, we are interested in the second method involving powerful EM pulses.

After recognizing the inherent threat of EMFI, extensive research efforts have been targeted in recent years to fully understand and mitigate its impact on diverse devices, ranging from microcontrollers to FPGAs and Application-Specific Integrated Circuits (ASICs). Our focus in this work will be directed towards FPGAs as they represent good candidates for implementing cryptographic algorithms and security primitives such as Physical Unclonable Functions (PUFs) and True Random Number Generators (TRNGs). Since these primitives employ ROs as an essential building block in their design, assessing their response to EMFI is important to improve their robustness.

Within this context, this paper aims to enhance the understanding of the harmonic locking impact on ROs with pulsed EMFI in FPGAs. This effect was introduced in [4] where the authors were able to lock the RO frequency into one of its harmonics through a single properly tuned EM pulse. The RO harmonic response was characterized as a function of its placement in the FPGA chip, the EM pulse width (PW) and amplitude, and the position of the probe relative to the FPGA package. In [5], the investigation sought to understand how multiple pulses and the placement and routing (P&R) constraints of both the inverters and the input/output (IO) pins influence the sensitivity of the ring. In this study, we advance from the groundwork of these papers and our recent work [16] to elaborate on the influence on the harmonic errors of other parameters related to the EM pulse timing and polarity, the manufacturing properties of the FPGA and the RO layout and density within the FPGA die. In more detail, the main contributions of this paper consist of:

- Assessing the impact of the FPGA structure (Spartan7 vs. Artix7-100T) and its type of packaging (Wire-bond vs. Flip-Chip) on the fault distribution and its magnitude on different modern FPGA platforms.
- Investigating the pulse polarity effect on faults to emphasize its significance when evaluating the efficiency of EMFI across various FPGAs.
- Investigating the influence of P&R constraints on maximizing the harmonic response while varying the density and the layout of the RO with a constant number of inverters.
- Highlighting the importance of optimal utilization of interconnects to enable an efficient RO-based detector with minimal overhead.
- Analyzing the injection timing impact on the control over the range of induced harmonic errors to verify the attacker's ability to circumvent the detectors.

The remainder of this paper is organized as follows. Section 2 introduces the related works on EMFI as well as the locking phenomenon on ROs. The experimental setup

and methodology for the tests are described in Section 3. Section 4 reports and analyzes the experimental results. Finally, Section 5 draws conclusions and perspectives.

2 Background

2.1 Electromagnetic Fault Injection

EMFI corrupts the normal operation of an integrated circuit (IC) through parasitic currents that are induced in all wire loops of the IC after a sudden variation of the magnetic or electric field generated by a probe. The EM coupling can be achieved through two different methods: harmonic EMFI and pulsed EMFI. The former consists of exposing the ICs to continuous EM waves to usually target analog blocks of ICs whose operation is continuous in time like clock generators and TRNGs as studied in [17] and [18]. On the other hand, pulsed EMFI disrupts the behavior of ICs during a few clock cycles by generating sudden variations of the magnetic field in a reduced volume close to the IC surface that induce parasitic currents in the closed loops of the power and ground networks of the DUT. In this study, we are mainly interested in single short perturbations; therefore, our EMFI attack will be focused on EM pulses.

It should be noted that mounting a successful EMFI attack includes multiple steps and identifying the optimal sensitive location on the die is the critical one. To achieve that, one can leverage EM side-channel to find the maximum leakage observed on the target chip. Within this context, a recent paper [6] investigated the design optimization of a hybrid EM probe that can efficiently enable both capturing EM emissions and injecting EM perturbations. This can help reduce the time and complexity of the evaluation that still includes other parameters related to the pulse such as the injection timing, and the width, amplitude, and polarity of the pulse.

2.2 Related Works

EM Fault Models & Countermeasures. Considering all these challenges, ensuring full protection of different devices against EMFI requires accurate fault models describing the mechanism involved in EM-induced faults. Several works [7] [8] [9] have been carried out to explain their nature. They suggested that these faults can be explained either by the Timing or by the Sampling fault model, depending on the clock frequency of the target and the strength of the EM coupling within the circuit. A recent study by Nabhan et al. [10] highlighted that two underlying mechanisms are involved within the timing fault model. The first one relates to the target's clock distribution network which becomes highly susceptible to voltage glitches on the clock tree at low or moderate frequencies. However, at high frequencies associated with small slack, the power distribution network (PDN) is the most sensitive on-chip network as EM perturbation leads to violations in timing constraints. In another paper [11] trying to uncover the complexity of the EM injection and its interaction with circuits at a low physical level, the authors conducted EMFI experiments from the substrate backside of a 180 nm CMOS technology equipped with six crypto cores of 128-bit AES and an on-chip voltage waveform monitor. Their results showed a key finding emphasizing that depending on

the polarity of the probe, significant negative voltage drops can be induced at specific locations within the PDN originating from injection positions that are not necessarily close to the impacted point.

Previous works studied the implementation of EMFI countermeasures on FPGAs such as [12] and [13]. However, the optimal number and location of the sensors to guarantee security against EM perturbations with the lowest overhead is still an open question. This primarily stems from the fact that the customization of placement and routing is challenging in FPGA design tools as they are designed by default to guarantee the best performance and resource optimization but do not take into account the circuit's EM emissions or susceptibility issues during the P&R process. To this end, our work will try to give more insights into the effects of P&R constraints on EMFI susceptibility to determine the best strategy for implementing detectors in different FPGA platforms.

Locking of Ring Oscillators. A RO is harmonically locked when it is forced to oscillate at a harmonic frequency (i.e., a multiple of the fundamental frequency). Previous studies on ROs focused mainly on the locking phenomenon happening due to radio-frequency interference on the power supply [19] or sinusoidal perturbation signals passed across a delay line placed near the RO [20], which makes them lock onto a signal with a frequency close to their natural oscillation frequency or its harmonics. This can render the confidential key generated using RO-based TRNGs partially or even fully deterministic by controlling the bias as demonstrated in [17] [21] through EM harmonic injections.

Sustained Harmonic Errors. This phenomenon was achieved with both laser experiments on a custom-designed 40nm Bulk CMOS ring oscillator, and electrical experiments on a RO constructed with discrete components in [14]. The authors were able to conclude that when one or several Single Event Transients (SETs) with a pulse width smaller than the total loop delay are induced during one oscillation period T of the RO, it deviates from its fundamental frequency and locks to one of its odd harmonics depending on the induced number of extra rising edges. In [4], the authors were able to validate this behavior through the injection of a single positive EM pulse into a RO composed of 1200 (50 x 24) inverters and implemented in an Artix7-100T FPGA. In their work, the ring was placed either on the top or bottom clock region and used 1 Look-Up Table (LUT) per Slice. The results showed that the placement of the RO in the chip, and the location and intensity of the injection affect the fault occurrence: when placed on the top clock region, the RO was more vulnerable to harmonic errors. Also, it was shown that in order to force higher harmonics with higher probabilities, one can increase the PW and amplitude of the pulse.

In [5], the characterization included a first analysis of the impact of routing constraints on the harmonic response of the RO. The findings suggested that for the same placement of the ring, favoring short vertical connections (called 'vertical snake') between LUTs results in higher harmonic sensitivity, while the lowest vulnerability was reported with long horizontal connections. Conducting similar tests on Spartan7 and Kintex7 FPGAs enabled the confirmation of this conclusion. Therefore, in this work,

we will adopt the most sensitive routing ‘vertical snake’ to investigate further parameters: how the pulse polarity, the density, the layout of the RO and the injection timing impact the fault distribution and the harmonics intensity on different FPGAs.

3 Experimental Setup and Methodology

3.1 EMFI Setup

The following setup shown in Fig. 1 was used to perform the EMFI experiments:

- Pulse Generator: We used the ChipShouter pulse generator for this work to perform EM pulsed fault injection. It can generate pulses with amplitudes from 150V up to 500V and variable pulse widths depending on the probe’s diameter and the voltage amplitude.
- XYZ Motorized Table: The motors are used to precisely control the position of the EM probe. They can set the X, Y, and Z positions via an RS-232 serial interface.
- EM Probes: We used two of the EM probes provided with the ChipShouter and consisting of a 1mm wire coiled either clockwise (CW) or counter-clockwise (CCW) around a 4mm ferrite core to induce positive or negative pulse polarity, respectively.
- Control PC: controls the whole platform through serial ports.
- Digital Oscilloscope: A Picoscope of 200 MHz bandwidth monitored the frequency of the RO and the synchronization between its oscillations and the pulse injection.
- Targets: Our characterization has been achieved on four types of AMD-Xilinx FPGAs all manufactured in 28nm process technology: a Spartan7 (xc7s25-1CSGA225) in a Cmod-S7 board; two Artix7 FPGAs (xc7a100T-1CSG324 & xc7a200T-1SBG484) within Nexys-A7 and Nexys-Video boards respectively, and finally a Kintex7 (xc7k160T-1FBG676) embedded in a SAKURA-X board. Their characteristics are summarized in Table 1.

It should be noted that each FPGA die covers only a portion of the package. The die sizes of Kintex7 and Artix7-200T were measured since both were decapsulated in their

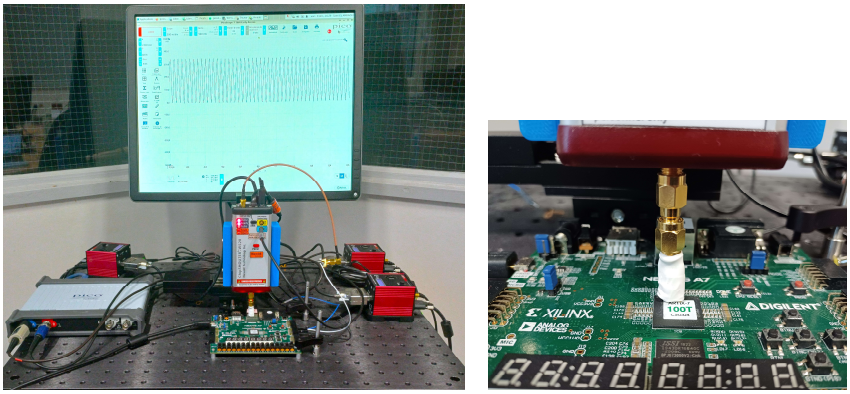


Fig. 1. Experimental setup

Table 1. Characteristics of the targeted FPGAs

FPGA	Packaging	Package Size (mm^2)	Die Size (mm^2)	Logic Slices
Spartan7	Wire-bond chip-scale	13 x 13	5 x 5.1	3650
Artix7-100T	Wire-bond chip-scale	15 x 15	6.5 x 10	15850
Artix7-200T	Flip-Chip lidless	19 x 19	10.66 x 11.95	33650
Kintex7	Flip-Chip lidless	27 x 27	9 x 12	25350

boards. For Spartan7, this information was revealed through X-ray imaging while it was reported in [15] for Artix7-100T. Both Spartan7 and Artix7-100T use a wire-bond packaging technology, where the die is attached to the substrate face up and the connections are made with wires. On the other hand, the Kintex7 and Artix7-200T are flipped over and placed face down, which eliminates wire bonds and results in lower inductance and better thermal performance. All these differences will be leveraged to explore how they will affect the outcome of EMFI.

3.2 RO Design

Ring Oscillator. The oscillating frequency of a RO depends on various parameters but its simplified computation can be done with the following formula, where D_i is the propagation delay of each delay element in the RO, D_r is the average delay related to the routing between them, and N is the number of these elements:

$$F = \frac{1}{2 \times (D_i + D_r) \times N}$$

Fig. 2 shows the architecture of our implemented RO with an even number N = 1200 of inverters and a Nand gate used as an activation gate. After the Enable signal moves from a logical low level to a logical high level, oscillations start. Then if we force the Enable signal to move low, the oscillations stop.

Placement and Routing Constraints. In all targeted FPGAs, the Configurable Logic Block (CLB) tile contains two slices. Each slice includes 4 LUTs and 8 flip-flops. Depending on the density that we want to achieve, our RO was formed by configuring either one or all of the 4 LUTs within each slice as an inverter. This makes the ring more or less compact in the FPGA. Also, to explore the effect of different RO layouts,

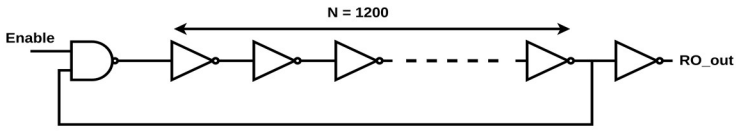


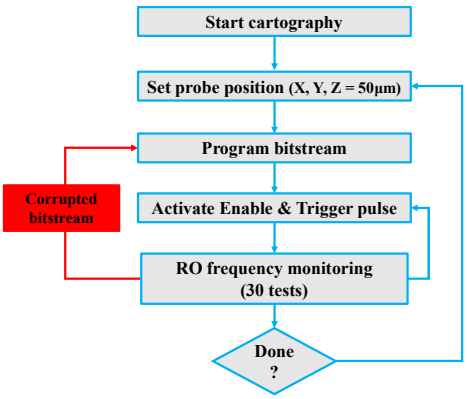
Fig. 2. Architecture of the ring oscillator

we customized the placement and routing of the LUTs in Vivado tool to enable either a horizontal, vertical, or square shape. It should be noted that in all our experiments, we used the same number of inverters (N=1200 to constrain the RO to one clock region) for each different P&R implementation of the RO in all three FPGAs.

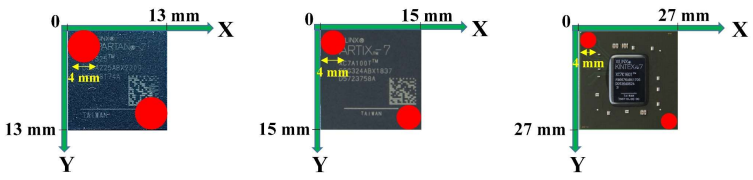
3.3 Methodology

After a single EM pulse injection, the RO frequency may change or not, as follows:

- **Unchanged frequency:** In that case, after the attack, the RO still oscillates at the same fundamental frequency. If we disable the RO and it keeps oscillating, we know that the 'Enable' configuration was corrupted, and the FPGA must be reprogrammed for the next test.
- **Harmonic locked frequency**: After the attack, the RO frequency can be locked into one of its odd harmonics (3, 5, 7….). Therefore, a reset of the 'Enable' is mandatory to restore the original oscillation frequency of the ring. However, as in the previous case, lack of control over the 'Enable' means that the board must be reprogrammed.
- **Noise signal**: the attack forces the RO output signal to noise, which means the bitstream was corrupted as resetting the enable signal does not restart oscillations. Therefore, reprogramming the bitstream is always mandatory before the next test.



(a)



(b)

Fig. 3. (a) RO faults evaluation flow (b) FPGA Package Scan

Based on these observed effects, a specific methodology, depicted in Fig. 3(a), was adopted to inject and observe a single pulse into the RO. The goal is to detect the occurrence of harmonics while scanning over the FPGA package with the 4mm CW or CCW probe. The probe tip is kept 50µm on top of the package and displaced by steps of 1 mm (due to the probe's size and resolution) from the top left to bottom right corner as depicted in Fig. 3(b). The detailed steps are:

- 1st step: Set the initial EM pulse parameters for the test (amplitude = 450V / PW = 100ns). The choice of these values is motivated by the results reported in [4].
- 2nd step: Place the probe at the initial coordinate [X = 0, Y = 0] above the package.
- 3rd step: Program the FPGA with the bitstream.
- 4th step: Enable the RO and trigger the single EM pulse injection.
- 5th step: Monitor the output RO frequency after injection then disable the RO to detect the occurrence of harmonic induced frequencies and bitstream corruptions.
- 6th step: Repeat 30 times the steps 4 and 5 to assess the reproducibility of faults for the same (X, Y) coordinate.
- 7th step: Move the probe to the next position and restart the procedure at step 3 with a displacement step of 1mm until the last coordinate to obtain a fault sensitivity map of the whole FPGA package.

4 Experimental Results and Discussion

The experimental results reported in this section show the effect of the EM pulse polarity and timing on the different targeted FPGAs along with the impact of the RO layout on faults.

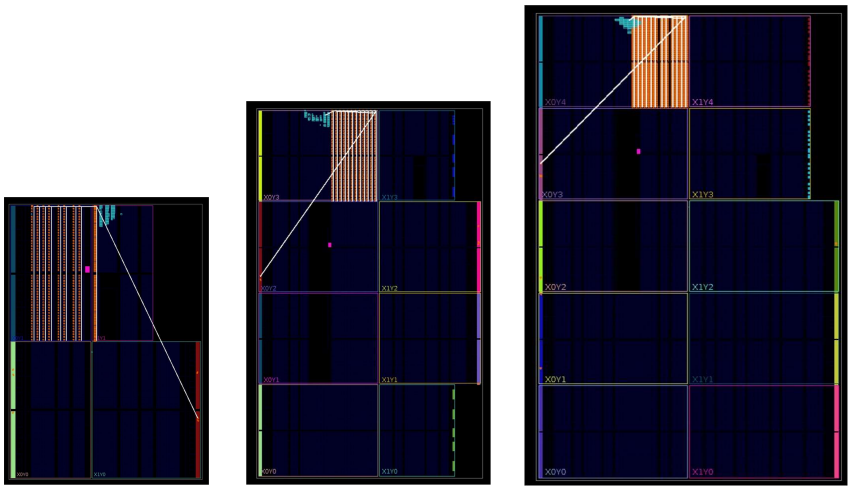
To improve the readability of the fault sensitivity maps, we assigned a specific color for each effect:

- **White**: No faults (Frequency remained the same and the bitstream was not corrupted).
- **Gradient from Yellow to Red**: shows the highest harmonic error that was monitored within 30 conducted tests.
- **Black:** represents mutes where reprogramming the bitstream was mandatory; the probability of bitstream corruption in this case is 100%.

It should be noted that the numbers in these maps refer to the ratio between the monitored frequency after EM injection and the fundamental frequency of the targeted RO.

4.1 Impact of the FPGA Structure and Packaging

To explore the influence of the FPGA structure and packaging on the fault sensitivity of the RO, we followed the methodology shown in Fig. 3(a) by conducting EM injection campaigns targeting the same RO implemented in the three FPGAs; 1 LUT/Slice and a vertical routing with short connections, as illustrated in Fig. 4, were used. It should be



(a) (b) (c)

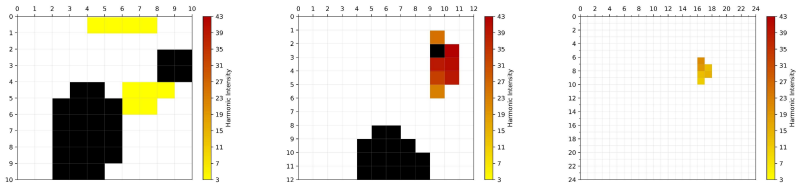
Fig. 4. FPGA floorplan from Vivado showing the implemented RO in (a) Spartan7 (b) Artix7-100T (c) Kintex7

noted that even if the same P&R of the RO was adopted, its frequency slightly varied between them because of the difference in their internal structure and the output pin connection. The RO was originally running at 1250 KHz in Spartan7, at 1075 KHz in Artix7-100T, and at 1417 KHz in Kintex7.

Fig. 5 represents the fault sensitivity maps of the three FPGAs with the positive polarity and shows that changing the FPGA family and packaging, even within the same process technology, can effectively influence the intensity and the location of the induced faults. The maximum impact was achieved in Artix7-100T with a harmonic error of 43, twice larger than the impact on Kintex7. Additionally, Spartan7 was the most sensitive to bitstream corruptions, as opposed to Kintex7 with no recorded vulnerability to this type of faults. This shows how Flip-Chip packaging, usually dedicated to high-performance FPGA, can reduce the fault sensitivity compared to wire-bond packaging. On another hand, if we focus on this later technology used for Spartan7 and Artix7-100T, by comparing Fig. 5(a) and Fig. 5(b) we notice some similarities in the location of faults especially in the bottom center coordinates. However, only third harmonic errors were induced for Spartan7, likely because it includes less programmable logic, hence a lower coupling between the probe and its PDN.

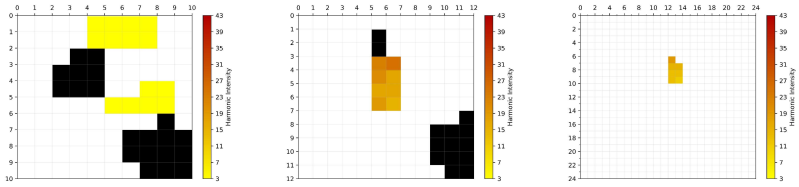
4.2 Impact of the Pulse Polarity

Similar tests were conducted to investigate the effect of changing the pulse polarity and the results are depicted in Fig. 6, showing the new fault locations in the three FPGAs. Upon an initial examination of the three cartographies, one may observe a similar trend



(a) (b) (c)

Fig. 5. Fault sensitivity maps with a **positive** pulse polarity (a) Spartan7 (b) Artix7-100T (c) Kintex7



(a) (b) (c)

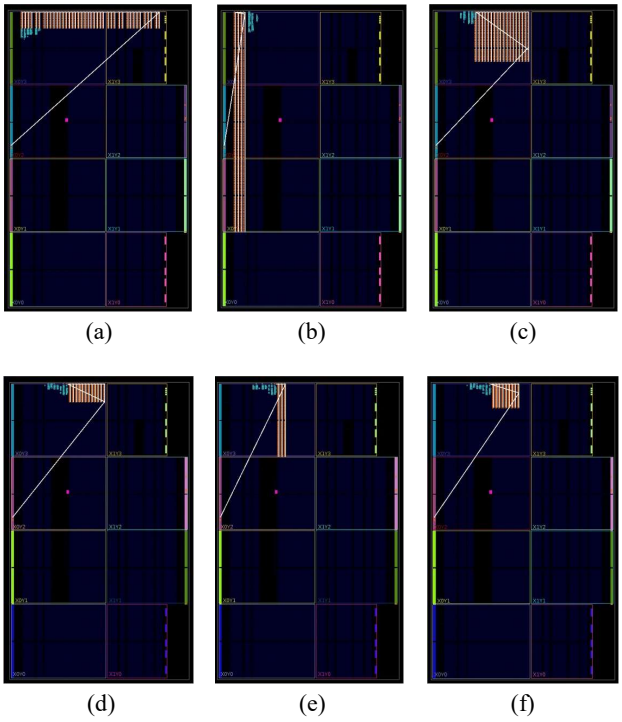
Fig. 6. Fault sensitivity maps with a **negative** pulse polarity (a) Spartan7 (b) Artix7-100T (c) Kintex7

compared to the positive polarity with Artix7-100T being the most vulnerable to harmonics and Spartan7 being the least. However, comparing Fig. 5 and Fig. 6 highlights that the negative polarity reveals complementary fault locations in the FPGA package. This indicates a different susceptibility of the power and the ground network within FPGAs to the polarity of EMFI, which underlines the importance of this parameter when assessing the robustness of a design against this attack.

4.3 Impact of the RO Layout

To further investigate the effect of changing the RO layout within the die, we performed new experiments while changing the placement constraints of the LUTs to enable different shapes of the RO. The intended goal is to obtain further insights into the impact that density and connections have on the ring sensitivity. Fig. 7(a) to (c) show the three obtained RO shapes (horizontal, vertical and square) achieved through the use of 1LUT/Slice. For the sake of conciseness, although tests were conducted on all FPGAs using positive and negative pulse polarities, we will here present only the results related to the Artix7-100T FPGA with negative polarity and then compare and discuss the other cases. Table 2 lists the frequency of the RO depending on its layout. As shown in this table, changing the layout results in a difference of less than 1% between the output frequencies: this is mainly due to the different use of horizontal and vertical connections.

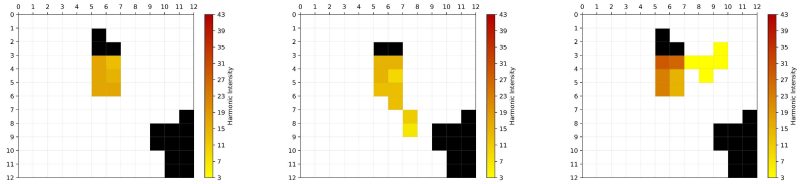
The results are represented in Fig. 8, showing how the fault sensitivity varies with the layout strategy. The most vulnerable layout was the square one with more sensitive locations and a harmonic error of 29. This indicates a better coupling between the probe and the FPGA when the RO is more compact within a single clock region, with few horizontal connections between the slices. This is likely due to the optimal density of inverters within the EM pulse. Similar tests on Spartan7 and Kintex7 enabled us to draw the same conclusion for both pulse polarities.



(a) (b) (c)

(d) (e) (f)

Fig. 7. Different layouts of the RO implemented in **Artix7-100T** FPGA using (above) **1LUT/Slice** (a) Horizontal (b) Vertical (c) Square and (below**) 4 LUTs/Slice** (d) Horizontal (e) Vertical (f) Square



(a) (b) (c)

Fig. 8. Fault sensitivity maps with a **negative polarity** using **1 LUT/Slice** configuration (a) Horizontal (b) Vertical (c) Square

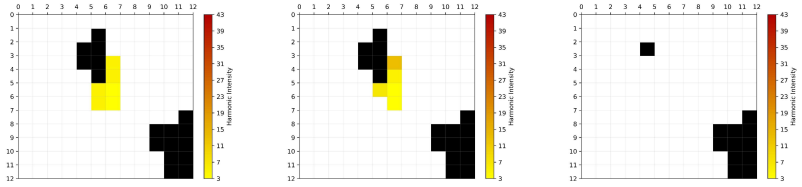
4.4 Impact of the RO Density

After showing that choices in the RO distribution within the chip lead to consistent differences in the experimental results between the three FPGAs, we aimed to highlight the impact of the RO density on the observed faults. To this end, we implemented similar configurations but using 4LUTs/Slice instead of only one as illustrated in Fig. 7(d) to (f). Their corresponding frequencies are detailed in Table 2.

The results reported in Fig. 9 show how minimizing the use of Slices (from 1200 down to 300) had a noticeable impact on the harmonics intensity compared to the results in Fig. 8, especially for the square layout where the harmonic sensitivity was completely suppressed. This was mainly caused by the low spatial resolution of the 4mm probe that cannot couple with elements of a small size, which indicates the likely need for a more advanced setup to induce harmonic errors. The same result was obtained for

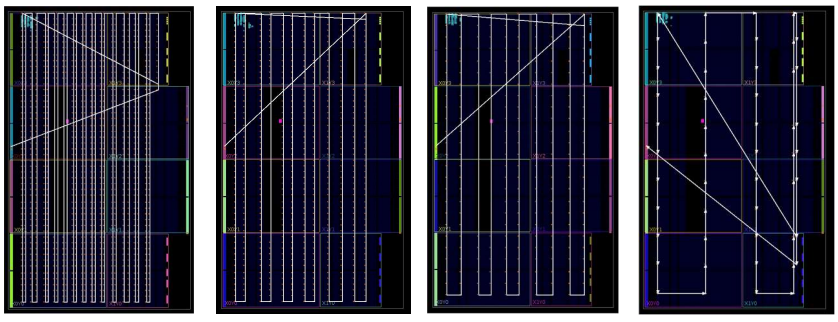
Table 2. Frequency of the RO with different layouts in the FPGAs

FPGA	Density	Layout	Frequency (KHz)
Artix7-100T	1 LUT/Slice	(a)	1086
		(b)	1076
		(c)	1078
	4 LUTs/Slice	(d)	1550
		(e)	1540
		(f)	1546



(a) (b) (c)

Fig. 9. Fault sensitivity maps with **negative polarity** using **4LUTs/Slice** configuration (a) Horizontal (b) Vertical (c) Square



(a) (b) (c) (d)

Fig. 10. RO implementation using 1LUT/Slice in **Artix7-100T** (a) N = 1200 (b) N = 600 (c) N = 228 (d) N = 54

the other FPGAs, which suggests that if we seek a minimal harmonic response of the RO, then adopting a square layout using 4 LUTs/Slice along with horizontal routing as reported in [5] remains the best option. This key finding is important for the robustness of RO-based PUFs and TRNGs against EMFI.

In the opposite direction, we tried to investigate the impact of making the RO layout less compact compared to the previous configurations while distributing the inverters all around the chip using 1LUT/Slice as depicted in Fig. 10. To explore the effect of reducing the number of delay elements on the harmonic errors, we implemented four RO implementations using 1200, 600, 228 and 54 inverters as shown in Fig. 10(a) to

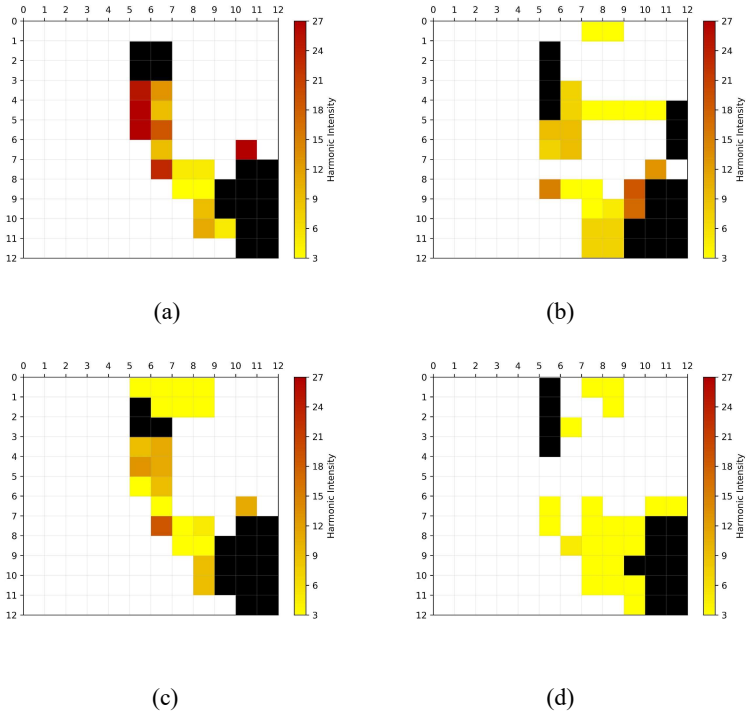


Fig. 11. Fault sensitivity maps with **negative polarity** (a) N = 1200 inverters (b) N = 600 inverters (c) N = 228 inverters (d) N = 54 inverters

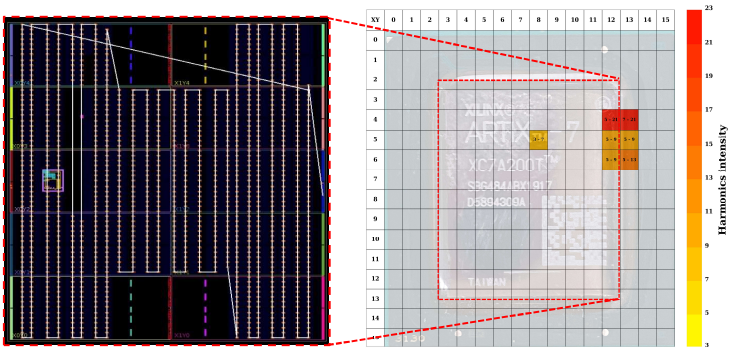
Table 3. RO characteristics across different configurations

Number of inverters	RO Frequency	Number of PIPs
1200	850 KHz	5549
600	1779 KHz	3150
228	4065 KHz	1334
54	11.7 MHz	478

10(d). Table 3 illustrates the RO frequency and the number of Programmable Interconnect Points (PIPs) for each configuration. The obtained fault maps are shown in Fig. 11 and they all demonstrate a higher number of impacted locations compared to previous results depicted in Fig. 8 and Fig. 9 where the RO was placed and routed within a specific region in the die. Furthermore, we can clearly observe that even when we reduce the number of inverters from 1200 down to 54 (factor of 22) which increases the frequency from 850 KHz to 11.7 MHz, we were still able to induce third and fifth harmonic errors on several locations. This indicates the importance of an optimal utilization of the PIPs in FPGAs for an effective EMFI detection with minimal overhead as they represent a critical block in the definition of the placement and routing constraints.

4.5 Injection Timing Effect

In the previous tests shown above, the attack scenario was based on the assumption that the attacker has low controllability over the timing of the attack: as a consequence, each performed injection induced the pulse at a different moment during the low or high level of the oscillation, thus injections were randomly spanning over the oscillation



(a) (b)

Fig. 12. (a) Floorplan of a RO implementation in **Artix7-200T** using N = 1200 inverters placed and routed throughout the whole die (Frequency = 921 KHz) (b) Fault sensitivity map using random EMFI with positive polarity (500V, 100ns)

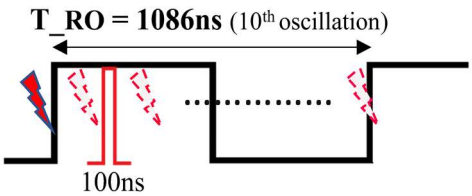
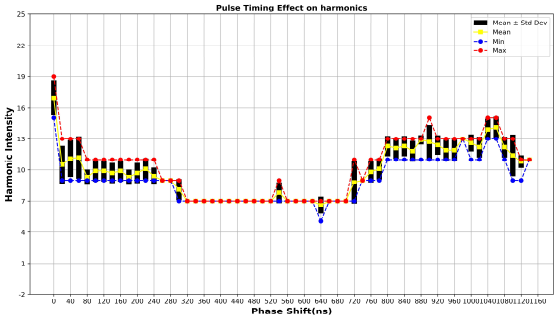
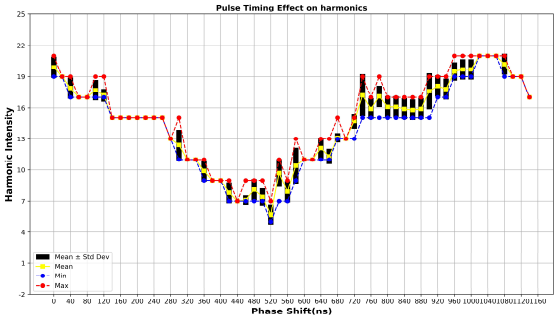


Fig. 13. Single EM pulse injection (PW=100ns) performed at different timings during the 10th oscillation period with a 20ns delay step configuration



(a)



(b)

Fig. 14. EM injection timing effect on harmonic errors (a) X12Y4 (b) X13Y4

period. However, another scenario might involve an attacker with high controllability over the timing of the attack. Therefore, to analyze the impact of the injection timing on harmonic errors, a state machine using UART communication was implemented to control the start of oscillations and to trigger a delayed EM pulse injection. When the RO oscillations are enabled, the FPGA outputs a trigger signal forcing the ChipShouter to precisely inject an EM pulse at the rising edge of the 10th oscillation. A delay step of 20ns is configured to systematically scan across the entire oscillation period; such granularity is based on our experimental setup, nonetheless it is much smaller than an oscillation period in order to produce meaningful results.

For the following tests, we chose to target an Artix7-200T FPGA embedded in a Nexys-Video board. Fig. 12(a) shows the RO implementation with 1200 inverters placed and routed throughout the whole die leading to an oscillation period of 1086 ns. Fig. 12(b) depicts the fault sensitivity map of the RO when targeted by random injections using a positive pulse of 500V and PW = 100ns. To enable a more detailed analysis over the evolution of the harmonics range with respect to the different injection timings, we focused on the most sensitive coordinates and conducted a campaign consisting of several experiments, each consisting of a single EM injection, with steps of

20ns to affect different phases of the oscillation period as depicted in Fig. 13. For each phase shift, the test was repeated 10 times to assess the fault reproducibility. It should be noted that although tests were conducted on all sensitive coordinates, only the harmonic errors variation at the X12Y4 and X13Y4 coordinates are presented in Fig. 14 as similar conclusions were drawn for the other positions. As a reminder, the initial timing delay of 0ns corresponds to a pulse injection synchronized with the rising edge of the 10th oscillation.

From the figure, we observe a periodic fluctuation of the harmonic errors for both coordinates. By selecting a specific injection timing during the oscillation period, one can be able to reduce or increase the harmonic susceptibility of the RO. Furthermore, it is important to highlight that the revealed errors for each specific delay represent subsets of the overall range obtained with a randomized timing. For instance, random injections on the X12Y4 position led to harmonic errors between 5 and 21 as shown in Fig. 12(b) compared to a (5 – 19) range with precise injection timing.

Therefore, we can deduce that tuning the injection timing parameter can be very efficient in controlling the outcome of an EMFI attack, which can be potentially leveraged, for example, to bias the response of RO-PUFs in FPGAs.

5 Conclusion

In this paper, we discussed the varying EMFI susceptibility of a RO while using various layouts in the FPGA. The evaluation on four FPGAs of different packaging and internal structures revealed the various challenges faced by both the attacker and the designer, which eventually led to the proposition of guidelines to consider for designing countermeasures that can effectively mitigate the EMFI threat in critical applications. By carefully analyzing the effect of pulse polarity on the harmonic response of the RO, we showed its significance when evaluating a detection mechanism to achieve an ideal behavior. Moreover, based on our analysis, we concluded that adopting a low compactness strategy when implementing the design of EMFI sensors can be advantageous for an optimal detection with minimal overhead. However, it should be noted that the attacker may still be able to circumvent the pulse detection when having a good control over the synchronization of the pulse injection. Therefore, it is mandatory to finely tune RO-based detectors in order to cover all possible attack scenarios.

As future work, we will try to leverage the key findings of this paper to assess and protect against the EMFI threat on RO-based PUFs or TRNGs implementations in FPGAs.

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